

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME:

Cloud Computing and
Big Data Analytics

COURSE CODE:

CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

I Deepak R(19251008) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Cloud Computing

Definition

A model for enabling ubiquitous convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Characteristics

- ⊙ On demand self service
- ⊙ Broad Network Access
- ⊙ Resource Pooling
- ⊙ Rapid Elasticity
- ⊙ Measured Service

Benefits

- ⊙ Cost Reduction
- ⊙ Scalability
- ⊙ Flexibility
- ⊙ High Availability
- ⊙ Recovery
- ⊙ Any where Access
- ⊙ Better Resource Utilization

Limitations

- ⊙ Privacy Concern
- ⊙ Internet Dependency
- ⊙ Vendor Lock-in
- ⊙ Downtime
- ⊙ Limited Control
- ⊙ Compliance Challenges

Hardware evolution

- ① 1934, Vacuum tubes } 1st Gen
- ① 1941 Turing complete
- ① 1946 ENIAC computer 2nd Gen
- ① 1971 Intel 4004 microprocessor 3rd Gen
- ① 1974 MITS Altair 8800 PC 4th Gen

Internet Software Evolution

- ① 1957 Sputnik, ARPA Established DARPA (1972)
- ① SAGE
- ① ARPANET (1967)
- ① TCP/IP
- ① IP v6 (1995)
- ① Webserver by Tim-Berners-Lee
- ① Mosaic Browser
- ① Mozilla (1994)

Evolution of Cloud Computing

Distributed system

① Cluster Computing



① Grid Computing



① Utility Computing



① Virtualization



① Cloud Computing

Service Delivery

- ① on Demand self Service
- ① Elastic Resource Provisioning
- ① Measured Service
- ① Global Access to service

IAAS	
Delivers virtualized infrastructure resource	
Provider Responsibilities	User Responsibilities
① Data Centers ① Physical Servers ① Storage ① Networking ① Virtualization ① Availability	① OS and Backups ① Middleware ① Runtime ① Application ① Data ① Security

PAAS	
Delivers Platform for application	
Provider Responsibilities	User Responsibilities
① Data Centers ① OS ① Storage ① Database ① Development Tools ① Networking ① Virtualization	① Application ① Data ① Security ① Backups ① Configuration settings

Service Models

SAAS	
Delivers complete Application over the Internet	
Provider Responsibilities	User Responsibilities
① Data Centers ① Physical Servers ① Storage ① Networking ① Virtualization ① Operating System ① Runtime ① Middleware	① Access the App ① Manages User ① Data input and Usage ① Security of own Data ① Configuration

XAAS	
Delivers "anything" as a service over the cloud	
Provider Responsibilities	User Responsibilities
① Data Centers ① Physical Server ① Storage ① Networking ① Virtualization ① OS / Middleware / Runtime ① Service specific Components ① Security ① Maintenance & Updates ① Availability	① Consume the Specific Service ① Configure ① Manage ① Security of own Data

Virtualization and web services as Enablers of Cloud

Virtualization

Provides

- ⊙ Abstraction of hardware
- ⊙ Multiple VM on single system / server
- ⊙ Resource pooling
- ⊙ Dynamic allocation of resources
- ⊙ Hardware independence

leads to

- ⊙ Efficient utilization of hardware
- ⊙ Reduced cost
- ⊙ Rapid provisioning of resource
- ⊙ Easy scaling

Cloud Deployment

supports

- ⊙ On demand Service
- ⊙ Elastic Resource
- ⊙ High Availability
- ⊙ Multi-tenancy
- ⊙ Pay as you go

result in

Scalability

Scale Up/out

Handle Variable Workloads

Improve performance

Support Growth and Expansion

Web Services

Provides

- ⊙ Standard communication
- ⊙ Platform independent
- ⊙ Loose coupling
- ⊙ Discoverability and interoperability
- ⊙ Service reuse

leads to

- ⊙ Easy integration of services
- ⊙ Reusability of components
- ⊙ Interoperable systems
- ⊙ Faster development